

Regulation of Hippo pathway transcription factor TEAD by p38 MAPK-induced cytoplasmic translocation

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Evidence abstract

The Hippo pathway controls organ size and tissue homeostasis, with deregulation leading to cancer.

The core Hippo components in mammals are composed of the upstream serine/threonine kinases Mst1/2, MAPK4Ks and Lats1/2.

Inactivation of these upstream kinases leads to dephosphorylation, stabilization, nuclear translocation and thus activation of the major functional transducers of the Hippo pathway, YAP and its paralogue TAZ.

YAP/TAZ are transcription co-activators that regulate gene expression primarily through interaction with the TEA domain DNA-binding family of transcription factors (TEAD).

The current paradigm for regulation of this pathway centres on phosphorylation-dependent nucleocytoplasmic shuttling of YAP/TAZ through a complex network of upstream components.

However, unlike other transcription factors, such as SMAD, NF- κ B, NFAT and STAT, the regulation of TEAD nucleocytoplasmic shuttling has been largely overlooked.

In the present study, we show that environmental stress promotes TEAD cytoplasmic translocation via p38 MAPK in a Hippo-independent manner.

Importantly, stress-induced TEAD inhibition predominates YAP-activating signals and selectively suppresses YAP-driven cancer cell growth.

Our data reveal a mechanism governing TEAD nucleocytoplasmic shuttling and show that TEAD localization is a critical determinant of Hippo signalling output.

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